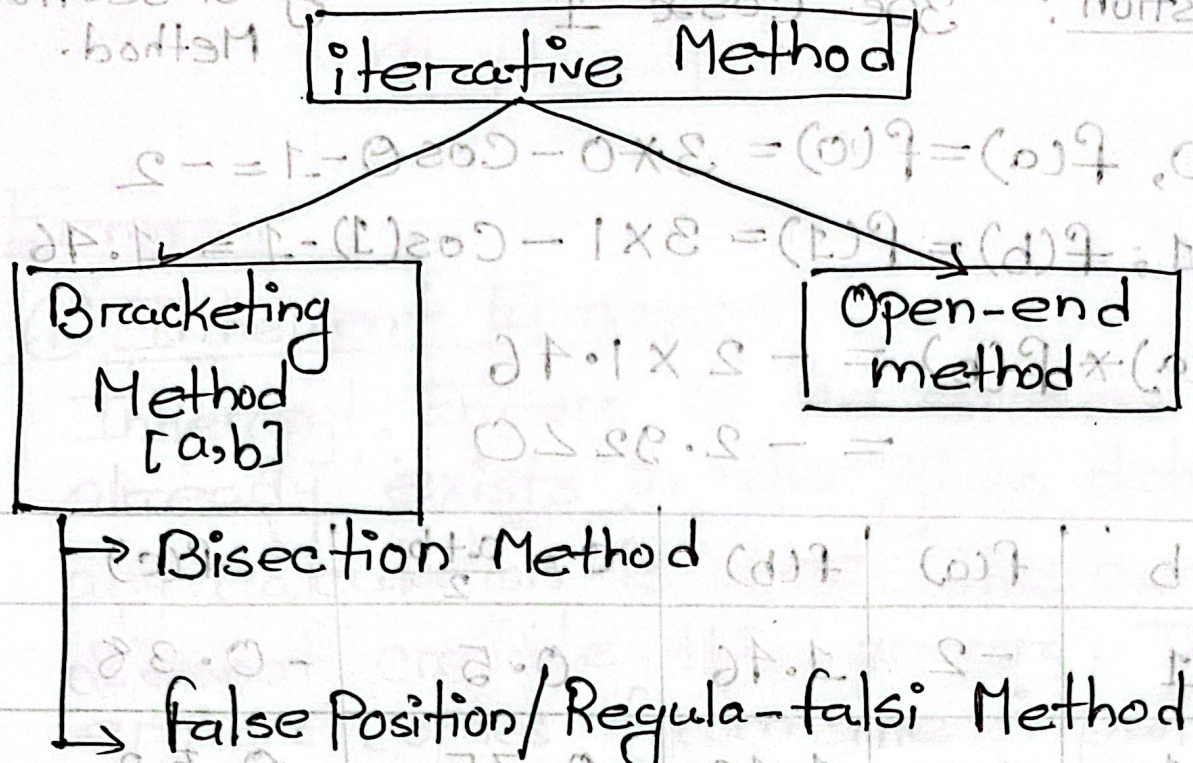
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Algorithm for Bi-section Method:-

1. Choose two real numbers a & b. such that $f(a) * f(b) < 0$.
2. define root, $c = \frac{a+b}{2}$
3. find $f(c)$.
4. if $f(a) * f(c) \leq 0$ then set $b = c$ else $a = c$

Return to step 1 until finding the root matched twice.

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Find the root from the following function using bi-section Method.

Question: $3x - \cos x - 1$

$\Rightarrow a = 0,\ f(a) = f(0) = 3*0 - \cos 0 - 1 = -2$

$b = 1;\ f(b) = f(1) = 3 \times 1 - \cos(1) - 1 = 1.46$

$\therefore f(a) * f(b) = -2 \times 1.46$

$= -2.92 < 0$

a	b	f(a)	f(b)	$c = \frac{a+b}{2}$	f(c)
0	1	-2	1.46	0.5	-0.38
0.5	1	-0.38	1.46	0.75	0.52
0.5	0.75	-0.38	0.52	0.625	0.06
0.5	0.625	-0.38	0.06	0.56	-0.16
0.56	0.625	-0.16	0.06	0.59	-0.05
0.59	0.625	-0.05	0.06	0.61	0.00
0.59	0.61	0.05	0.00	0.60	-0.02
0.60	0.61	-0.02	0.00	0.61	-0.01

Root = 0.61

$f(a) * f(c) \rightarrow$ Positive

$a = c$

$b = c$

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Root Calculation

~~Modelling Errors~~:

False-Position/Regula-falsi Method:

Algorithm:

1. Choose two real numbers a and b such that $f(a)*f(b)<0$
2. Define root, $c = \{a*f(b) - b*f(a)\} / \{f(b) - f(a)\}$
3. find $f(c)$.
4. If $f(a)*f(c) \leq 0$ then set $b=c$ else $a=c$
5. Return to step 1 until finding the root matched twice.

Example:

$f(x) = 3x - \cos x - 1$

$a=0;\ f(a) = f(0) = 3*0 - \cos(0) - 1 = -2$

$b=1;\ f(b) = f(1) = 3*1 - \cos(1) - 1 = 1.46$

$\therefore f(a)*f(b) = -2*1.46$

$= -2.92 < 0$

$a=0,\ b=1$

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Itr	a	b	f(a)	f(b)	c	f(c)
0	0	1	−2	1.46	0.578	−0.103
1	0.578	1	-0.103	1.46	0.606	−0.004
2	0.606	1	-0.004	1.46	0.607	−0.0004
3	0.607	1	−0.0004	1.46	0.607	−0.0004

Root = 0.607.

Fixed point Method:-

Algorithm:-

1. Given an equation in $f(x)=0$
2. Convert $f(x)=0$ into the form of
$$x = g(x).$$
3. Let the initial guess ($\log x$, $\sin x$, $\cos x$, $e^x = 3.2$)
→ Simple = (0.5)

Do,
$$x_{i+1} = g(x_i)$$
while (none of the convergence iteration is matched).

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Ex: $f(x) = 3x - \cos x - 1$

Let, $f(x) = 0$

$\Rightarrow 3x - \cos x - 1 = 0$

$\Rightarrow 3x = \cos x + 1$

$\Rightarrow x = \frac{1+\cos x}{3}$, where $x = g(x)$,

$g(x) = \frac{1+\cos x}{3}$

Let, $x_0 = g(3.2)$

$= \frac{1+\cos(3.2)}{3}$

$= 0.0005 \approx 0$

Iteration:

1) $x_1 = g(x_0) = g(0) = \frac{1+\cos(0)}{3} = 0.67$

2) $x_2 = g(x_1) = g(0.67) = \frac{1+\cos(0.67)}{3} = 0.59$

3) $x_3 = g(x_2) = g(0.59) = \frac{1+\cos(0.59)}{3} = 0.61$

4) $x_4 = g(x_3) = g(0.61) = \frac{1+\cos(0.61)}{3} = 0.61$

$\therefore$ root, c_3 is $= 0.61$

Ans.

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Newton-Raphson Method:-

Algorithm:-

1) Find $f'(x_n)$.
2) Find a, b so that $f(a) * f(b) < 0$
3) Assume, $x_0 = a$.
4) Find out root, $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
5) Find $x_1, x_2, x_3 \ldots\ldots x_n$ untill any two successive values are equal.

Ex: $f(x) = 3x - \cos x - 1$

$f'(x) = 3 - (-\sin x)$

$= 3 + \sin x$

$f(a) = f(0) = 3*0 - \cos(0) - 1 = -2$

$f(b) = f(1) = 3*1 - \cos(1) - 1 = 1.46$

$f(a) * f(b) < 0$

$\Rightarrow -2 * 1.46 < 0$

$\Rightarrow -2.92 < 0$

$\therefore a = 0, \quad x_0 = 0$

$b = 1.$

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$f'(0) = 3$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \Rightarrow x_0 = - \frac{f(x_0)}{f'(x_0)}$$

$$\Rightarrow x_{0+1} = x_1 \Rightarrow x_0 - \frac{f(0)}{f'(0)}$$

$$= 0 - \frac{-2}{3}$$

$$= \frac{2}{3}$$

$$= 0.673$$

x_n	$f(x_n)$	$f'(x_n)$	↓ Root $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
0	-2	3	$0 - \frac{-2}{3} = 0.67$
0.67	0.23	3.62	$0.67 - \frac{0.23}{3.62} = 0.61$
0.61	0.01	3.57	$0.61 - \frac{0.01}{3.57} = 0.61$

Root = 0.61

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Extra

Bi-section Method

1) $f(x) = x^3 - 4x - 9$

$a = 2, f(a) = f(2) = 2^3 - 4 \times 2 - 9 = -9$

$b = 3, f(b) = f(3) = 3^3 - 4 \times 3 - 9 = 6$

$\therefore f(a) * f(b) = -9 \times 6$

$= -54 < 0$

$\therefore a = 2, b = 3$

Step	a	b	f(a)	f(b)	$c = \frac{a+b}{2}$	f(c)
1	2	3	-9	6	2.5	-3.375
2	2.5	3	-3.375	6	2.75	0.796
3	2.5	2.75	-3.375	0.7968	2.625	-1.412
4	2.625	2.75	-1.4121	0.7968	2.6875	-0.339
5	2.6875	2.75	-0.3391	0.7968	2.7187	0.2209
6	2.6875	2.7187	-0.3391	0.2209	2.7031	-0.061077
7	2.7031	2.7187	-0.0610	0.2209	2.7109	0.0799
8	2.7031	2.7109	-0.0610	0.0799	2.7070	0.0093
9	2.7031	2.7070	0.0610	0.0093	2.7050	-0.0258
10	2.7050	2.7070	-0.0258	0.0093	2.7060	-0.0082

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∴ Approximate root, c = 2.706.

Ans.

2) $f(x) = x^3 - 4$

$a = 1, f(a) = f(1) = 1^3 - 4 = -3$

$b = 2, f(b) = f(2) = 2^3 - 4 = 4$

$f(a) * f(b) = -3 \times 4 = -12 < 0$

∴ a = 1, b = 2.

Step	a	b	f(a)	f(b)	$c = \frac{a+b}{2}$	f(c)
1	1	2	-3	4	1.5	-0.625
2	1.5	2	-0.625	4	1.75	1.359

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False-Position Examples:

1) $f(x) = x^3 - 4x - 9$ $\qquad c = \dfrac{a*f(b) - b*f(a)}{f(b) - f(a)}$

আগে করা $a = 2,\ b = 3$

Step	a	b	f(a)	f(b)	c	f(c)
1	2	3	-9	6	2.6	-1.824
2	2.8	3	-1.824	6	2.69	-0.295
3	2.69	3	-0.295	6	2.70	-0.117
4	2.70	3	-0.117	6	2.70	-0.117

Root = 2.70

② $f(x) = x^3 - 1$

$\Rightarrow a = 1,\ b = 2$

Step	a	b	f(a)	f(b)	c	f(c)
1	1	2	-3	4	1.428	-1.088
2	1.428	2	-1.088	4	1.551	-0.269
3	1.551	2	-0.269	4	1.579	-0.068
4	1.579	2	0.063	4	1.585	-0.018
5	1.585	2	-0.018	4	1.587	-0.0030
6	1.585	2	-0.003	4	1.587	-0.0030

Root = 1.587.

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Fixed / One-point Method:- (Ex)

(1) $f(x) = 2x^3 - 2x - 5$.

(1.) $f(x) = 0$

$\Rightarrow 2x^3 - 2x - 5 = 0$

$\Rightarrow 2x^3 = 2x + 5$

$\Rightarrow x^3 = \frac{2x+5}{2}$

$\therefore x = \sqrt[3]{\frac{2x+5}{2}}$

(2.) $x = g(x)$ where $g(x) = \left(\frac{2x+5}{2}\right)^{\frac{1}{3}}$

(3.) $x_0 = g(0.5) = \left(\frac{2\times 0.5+5}{2}\right)^{\frac{1}{3}} = 1.44$

$x_1 = g(x_0) = g(1.44) = \left(\frac{2\times 1.44+5}{2}\right)^{\frac{1}{3}} = 1.58$

$x_2 = g(x_1) = g(1.58) = \left(\frac{2\times 1.58+5}{2}\right)^{\frac{1}{3}} = 1.59$

$x_3 = g(x_2) = g(1.59) = \left(\frac{2\times 1.59+5}{2}\right)^{\frac{1}{3}} = 1.59$

Root = 1.59

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(2) $f(x) = x^2 - x - 1$

$\Rightarrow$ 1. $f(x) = 0$

$\Rightarrow x^2 - x - 1 = 0$

$\rightarrow x^2 = x + 1$

$\Rightarrow x = \sqrt[2]{x+1}$

2. $gx = g(x)$ where $g(x) = \sqrt{x+1}$

3. $x_0 = g(0.5) = \sqrt{0.5+1} = 1.23$

$x_1 = g(1.23) = \sqrt{1.23+1} = 1.49$

$x_2 = g(1.49) = \sqrt{1.49+1} = 1.58$

$x_3 = g(1.58) = \sqrt{1.58+1} = 1.60$

$x_4 = g(1.60) = \sqrt{1.60+1} = 1.61$

$x_5 = g(1.61) = \sqrt{1.61+1} = 1.61$

$\therefore$ root $= 1.61$.

(3) $f(x) = x^3 - 4x - 9$ (Ans = 2.70)

(4) $f(x) = x^3 - 4x$

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Newton Raphson (Examples):

1. $f(x) = x^3 - 4x - 9$.

$\Rightarrow$ 1. $f'(x) = 3x^2 - 4$

2. $a(a) = 2, f(a) = f(2) = 2^3 - 4\times2 - 9 = -9$

$b = 3, f(b) = f(3) = 3^3 - 4\times3 - 9 = 6$

$f(a) * f(b) < 0$

$\Rightarrow -9 * 6 < 0$

$= -54 < 0$

3. $x_0 = a = 2$

Step	x_n	$f(x_n)$	$f'(x_n)$	$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
0	2	-9	8	$x_{0+1} = 2 - \frac{-9}{8} = 3.125$
1	3.125	0.02	25.29	$x_2 = 3.125 - \frac{0.02}{25.29} = 2.76$
2	2.76	0.98	18.85	$x_3 = 2.76 - \frac{0.98}{18.85} = 2.70$
3	2.70	-0.117	17.87	$x_4 = 2.70 - \frac{-0.117}{17.87} = 2.70$

Root = 2.70

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(2) $f(x) = x^3 - 4$

$\Rightarrow$ Step (1) $f'(x) = 3x^2$

(2) $\Rightarrow$ আগের $a = 1, f(a) = -3$

$b = 2, f(b) = 4$

$f(a) * f(b) < 0$

$\Rightarrow -12 < 0$

(3) $\because x_0 = a = 1$

Step	x_n	$f(x_n)$	$f'(x_n)$	$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
0	1	-3	3	$x_1 = 1 - \frac{-3}{3} = 2$
1	2	4	12	$x_2 = 2 - \frac{4}{12} = 1.67$
2	1.67	0.66	8.36	$x_3 = 1.67 - \frac{0.66}{8.36} = 1.59$
3	1.59	0.019	7.58	$x_4 = 1.59 - \frac{0.019}{7.58} = 1.59$

Root = 1.59

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